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Role of oxides of nitrogen in the ozone-cardiorespiratory visit association☆

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ABSTRACT

Ozone (O_3)-induced health effects vary in terms of severity, from deterioration of lung function and hospitalization to death. Several studies have reported a linear increase in health risks after O_3 exposure. However, current evidence suggests a non-linear U- and J-shaped concentration-response (C-R) function. The potential increasing risks with decreasing O_3 concentrations may seem counterintuitive from the traditional standpoint that decreasing exposure should lead to decreasing health risks. Tus, the question of whether the increasing risks with decreasing concentrations are truly O_3-induced or might be from other C-R mechanisms. If these potential risks were not accounted for, this may have contributed to the risks observed at the low ozone concentration range. In this study, we examined the short-term effects of photochemical oxidant (O_x, parts per billiion) on outpatient cardiorespiratory visits in 21 Japanese cities after adjusting for other air pollutant-specific C-R functions. Daily cardiorespiratory visits from January 1, 2014 to December 31, 2016 were obtained from the Japanese Medical Data Center Co. Ltd. Similar period of meteorological and air pollution variables were obtained from relevant data sources. We utilized a time-stratified case crossover design coupled with the generalized additive mixed model (TSCC-GAMM) to estimate the association between O_x and cardiorespiratory outpatient visits, after adjusting for several covariates. A total of 2,588,930 visits were recorded across the study period, with a mean of 111.87 and a standard deviation of 138.75. The results revealed that crude O_x-cardiorespiratory visits exhibited a U-shaped pattern. However, adjustment of the oxides of nitrogen, particularly nitrogen monoxide (NO), attenuated the lower risk curve and subsequently altered the shape of the C-R function, with a substantial reduction observed during winter. NO- and nitrogen dioxide (NO_2)-adjusted O_x-cardiorespiratory associations increased nearly linearly, without an apparent threshold. Current evidence suggests the importance of adjusting the oxides of nitrogen in estimating the O_x C-R risk functions.

1. Introduction

Ozone (O_3)-induced health effects vary in terms of severity, from lung function deterioration, and hospitalization to death (Nuvolone et al., 2018). These health risks, in turn, translate into a disease burden in terms of morbidity and mortality. Malley et al. (2017) noted a global estimate of 1.04–1.23 million respiratory deaths attributable to O_3 exposure, whereas Anenberg et al. (2018) estimated that 9–23 million annual asthma emergency room visits globally in 2015 were attributed to O_3. In Japan, country-led air pollution reduction efforts have been instrumental in the decline of annual concentrations of particulate matter with 2.5 μm or less in diameter ($PM_{2.5}$); however, annual O_3 concentrations have remained mostly unchanged with annual maximum 1-h daytime O_3 concentrations hovering between 109.9 and 118.5 parts per billion (ppb) from 1990 to 2018 (Ito et al., 2021). These unchanged concentration range elicit an array of short- and long-term toxicological

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and epidemiological effects in the exposed population.

Several epidemiological studies have revealed both short- and long-term positive associations between O_3 and human mortality, which is a result of the oxidative stress caused by O_3 exposure (Hulo et al., 2011; Sunil et al., 2013). In Japan, Ng et al. (2013) noticed a statistically significant short-term O_3 effect on all-cause mortality (0.69%; 95% CI: 0.27–1.10) during the intermediate season between summer and winter. In addition to mortality, several studies have documented both short-term and long-term effects on morbidity outcomes such as hospitalizations (Lin et al., 2008; Luong et al., 2018) and emergency room visits (Carlsen et al., 2013; Tian et al., 2018).

Health risks associated with exposure to O_3 are largely driven by underlying physiological mechanisms, which are usually illustrated through concentration-response (C-R) curves. However, studies regarding O_3-health outcome C-R curves have been largely inconsistent, reporting a combination of linear (Chen et al., 2017) and non-linear C-R (Bae et al., 2015; Faustini et al., 2019; Huang et al., 2019) functional shapes. The motivation behind the linear fit is the conservative assumption of linearity in the first instance, driven by data, with further optimization whenever the data warrants a non-linear treatment. Studies utilizing a linear fit have mostly observed an increasing risk with increasing concentrations (Bell et al., 2004; Gryparis et al., 2004; Vicedo-Cabrera et al., 2020). In several instances, O_3-health outcome associations potentially exhibit thresholds, with risks increasing beyond a set point concentration (Kim et al., 2004). However, several current studies suggest that there are increasing health risks to the left and right sides of the set point concentration (where risks are assumed to be the lowest), which are apparent in either J- or U-shaped C-R curves (Bae et al., 2015; Faustini et al., 2019; Huang et al., 2019). The increasing risk with decreasing O_3 concentration, based on a J- and U-shaped C-R curve, may seem counterintuitive from the standpoint of exposure science because decreasing exposure is tantamount to decreasing health risk. However, there is a plausible mechanism that entails photochemical processes involving nitrogen oxides, namely nitrogen dioxide (NO_2) and nitrogen monoxide (NO), and their role in the titration of O_3 (Han et al., 2011; Song et al., 2011). Several studies have noted that the adjustment of nitrogen oxides lead to a reduction in O_3-related health risks (Faustini et al., 2019; Tao et al., 2012).

Against this backdrop, we sought to understand this phenomenon by determining the short-term effects of O_3 on cardiorespiratory visits, particularly considering the potential role of nitrogen oxides. Owing to the seasonal variation in O_3, we further examined the season-specific associations.

2. Methods

2.1. Study areas

The 21 study areas are located across the country as shown in Fig. 1. All cities, excluding Tokyo, are ordinance-designated cities (ODC) established by the Japanese government. ODC cities are characterized by 1) population size greater than 500,000, 2) a population density of greater than 2000 population per km^2, 3) being considered as a major city in their region, and 4) having less than 10% of their employment in primary industries (Ohsugi, 2011; Steiner, 1965). ODCs differ from ordinary cities because they can execute functions that are originally performed or delegated to prefectural governments (Jacobs, 2011).

2.2. Health outcome data

Daily cardiorespiratory visits of 21 cities from January 1, 2014 to December 31, 2016 were obtained from the Japanese Medical Data Center (JMDC) Co. Ltd. The anonymized information was based on the health insurance claims of the specific treatments and other demographic variables for employees of Japanese companies and their familiees, most of them were aged 65 years or younger (Kimura et al., 2010). Cardiorespiratory visits were coded using the International Classification of Disease (ICD) 10 for cardiovascular (I00–I99) and respiratory (J00-J99) diseases. Cardiorespiratory outpatient visits in July, August, and September were categorized as summer-related visits, whereas winter months included December, January, and February. Hereafter, we refer to these visits as cardiorespiratory outpatient visits, unless otherwise stated. This study was approved by the Ethics Committee of the Graduate School of Engineering, Kyoto University with the Institutional Review Board (IRB) No. 201714.

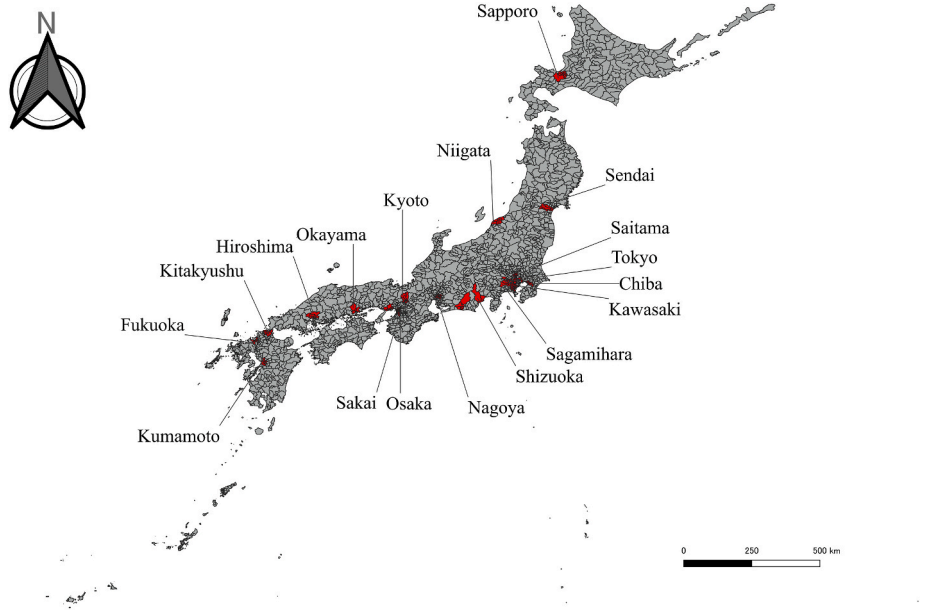


Fig. 1. Geographical location of the 21 cities included in the study, Study sites are distributed across the country from the Northernmost city of Sapporo down to the southern city of Fukuoka. Cities are indicated in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.3. Air pollution data

Air pollution data from January 1, 2014 to December 31, 2016, were obtained from the National Institute for Environmental Studies (NIES). Hourly air pollution data were obtained from background monitoring stations located within each city. Background monitoring stations with continuous and non-missing air pollution data were used in this study. The number of eligible background monitoring stations per city is presented in Table S1. For locations with more than one eligible background monitoring station, the mean across these locations was used. We took the 24-h mean concentration of NO (ppb; parts per billion), NO_2 (ppb) and $PM_{2.5}$ ($\mu g/m^3$). When monitoring stations with more than four out of the 24 missing data points, these were assumed to be missing data days. For O_3, we used photochemical oxidant (O_x) (ppb). In Japan, because O_3 is the major component of O_x, the Japanese Ministry of Environment decided to equate O_3 concentrations with O_x concentrations in 1996 (Fukunaga et al., 2021). Several studies have also used O_x in lieu of $O_3$when examining ozone-related effects on health (Chen et al., 2014; Ng et al., 2013; Vicedo-Cabrera et al., 2020). Ox is defined as an oxidizing substance produced mainly through photochemical reactions based on the potassium iodide (KI) method, a wet chemical method that responds to photochemical oxidants, primarily ozone ($\geq$90%), followed by peroxyacetyl nitrate (PAN), hydrogen peroxide (H_2O_2), and organic hydroperoxides (Akimoto, 2017).

Specifically, we utilized the maximum 8-hour moving mean ozone of the day. Days with missing values were interpolated utilizing the information of the neighboring days' data via the k-nearest neighbor interpolation (Daberdaku et al., 2020).

2.4. Meteorological data

Daily mean ambient air temperature (°C) and relative humidity (%) during the same study period were obtained from NIES. The summary statistics for the air pollution and meteorological variables are shown in Table S1.

2.5. Statistical analyses

We utilized a time-stratified case crossover design coupled with the generalized additive mixed model (TSCC-GAMM) to estimate the association between ozone and cardiorespiratory outpatient visits in the 21 cities after adjusting for the relevant covariates (as shown in Equation (1)).

Briefly, a case-crossover (CCO) study is a self-matched case-control study as proposed by Maclure (1991), wherein the exposure at each event is matched with the exposures for the same subject at his/her referent times. This is further expanded by Lumley and Levy (2000) to a time-stratified CCO, which takes referent times within the same stratum (i.e., month and day of the week) (Carracedo-Martínez et al., 2010). A more detailed specification of the study design is discussed elsewhere (Carracedo-Martínez et al., 2010; Janes et al., 2005; Lumley and Levy, 2000).

$$y_{i,t} \sim \text{Quasipoisson} \tag{1}$$

$$y_{i,t} = \beta_0 + ns(\text{ozone}_t, 3) + ns(airpol_t, df) + \beta_1 hod_t + \beta_2 stratum_t + influenza_t + ns\left(temp_{Lag0-4}, 5\right) + ns(RHave_t, 3) + random(1|city)$$

The daily cardiorespiratory outpatient visits ($y_{i,t}$) of city (i) in time (t) followed a Poisson distribution with a logarithmic link function. A quasi-Poisson distribution was used for overdispersion in the health outcomes. The outpatient visit ($y_{i,t}$) was modeled with the intercept (β_0), linear terms for holidays (hod_t), and binary influenza epidemic term ($influenza_t$), whereby we coded weeks (and eventually converted to days) by taking the influenza count over the study period exceeding 80% as 1, otherwise, 0 (Wang et al., 2017). The stratum of year, month, and day of the week ($stratum_t$) was used to represent the time-stratification within the case-crossover design. Non-linear terms included the natural cubic spline for same-day ozone, $ns(\text{ozone}, 3)$, with three degrees of freedom (df), 3-day lagged mean air temperature with 5df, and 3df for relative humidity. The oxides of nitrogen, NO and NO_2, were further adjusted in the model via the $ns(airpol_t, df)$, with 4 df and 3 df (based on the sensitivity analysis in Table S2), respectively. In subsequent sections, we draw comparisons among the three models, namely the ozone-only (crude), NO-adjusted, and NO_2-adjusted models. Since TSCC-GAMM or mixed models, in general, generated a pooled estimate of the 21 cities (Spieler and Schumacher, 2019), the succeeding results presented in this study are for the pooled association.

Based on our initial analysis, we observed a non-linear association between ozone concentration and visits. Thus, we defined the minimum risk point (MRP) as the concentration at which the risk of a cardiorespiratory visit was the lowest. Further preliminary results also revealed that accounting for the oxides of nitrogen yielded a candidate threshold model for the ozone-cardiorespiratory visit association. To ascertain whether a threshold existed, we followed the threshold analysis implemented in previous studies (Bell et al., 2006; Chen et al., 2017). In brief, the threshold analysis assumes that the ozone-cardiorespiratory risk below a potential threshold is null, whereas risks beyond (the potential threshold) are amonotonically increasing. We performed a potential threshold analysis for each city at each potential threshold over a range of concentrations from 0 to 80 ppb in 1 ppb increments. For each potential city and potential threshold, we estimated model performance using the Akaike Information Criterion for quasi-Poisson models (qAIC) (Akaike, 1974). We then calculated the mean of the qAIC values per potential threshold (Bell et al., 2006). The rationale of this approach is to simply compare the mean qAIC when the potential threshold is equal to zero, equivalent to the no-threshold model, with the lowest potential threshold across the range of potential thresholds. If a threshold exists, the mean qAIC will be lower than that of the no-threshold model. A lower qAIC value indicates a better fit. Specifically, if the difference between the potential threshold at 0 and the lowest potential threshold is more than two, there is likely to be a difference in the model fit (Burnham and Anderson, 2004; Chen et al., 2017). We further adjusted for the non-linear terms of same-day NO and NO_2 with 4df and 3df, respectively. The df specifications for the non-linear terms included in the model are summarized in Table S2.

In view of the seasonal variability of ozone, we also conducted a season-specific analysis in summer and winter. In addition, we used the 1st and 99th percentiles of the C-R curve to illustrate the changes below and above the MRP, respectively. Similarly, we utilized the season-specific 1st and 99th percentiles in determining the season-specific risks below and above the MRP. All analyses were performed using R Statistical Programming version 4.2.2 (R Core Team, 2019).

3. Results

3.1. Descriptive statistics

A total of 2,588,930 visits were recorded across 21 cities during the study period, with a mean of 111.87 and a standard deviation of 138.75 (Table 1). The 21 Japanese cities are located across the country, including the Sapporo (in the north) and the southern city of Kumamoto, highlighted in Fig. 1. Northern cities are relatively colder with Sapporo recording a year-round daily mean air temperature of 9.55 °C compared to southern cities, which are approximately 7 °C higher (Table S1). Air pollution concentrations are well below the daily national standards. Year-round, 24-h mean NO_2 concentrations as well as the mean 8-h maximum O_x during the study period are below the national daily standards of 40–60 ppb (for NO_2) and 60 ppb (for O_x), respectively. A year-round correlational association revealed a statistically weak negative correlation between O_x and cardiorespiratory visits (r = -0.05), whereas a moderately-weak and positive association was observed with

Table 1
Descriptive statistics for daily cardiorespiratory clinic visits, meteorological and air pollution variables.

	Mean	Standard deviation	Minimum	P25[a]	P50[a]	P75[a]	Maximum
Cardiorespiratory clinic visit	111.87	138.75	0	26	51	140	892
O_x (ppb)	41.83	15.05	2.44	31.59	39.84	51.08	122.78
NO (ppb)	3.47	4.87	0	1.08	1.93	3.82	73.14
NO_2 (ppb)	12.72	6.84	1.18	7.65	11.41	16.35	56.61
Mean temperature (°C)	16.06	8.2	−9.2	8.8	16.8	22.9	32.6
Relative Humidity (%)	67.84	13.78	19	59	68	78	100

O_x: photochemical oxidant (ozone); NO: nitric oxide; NO_2: nitrogen dioxide; ppb: parts per billion; μg/m^3: microgram per cubic meter; °C: degrees Celsius; %: percent.
[a] P25: 25th percentile; P50: 50th percentile; P75: 75th percentile.

either NO or NO_2 ($r = 0.16$) (Fig S1).

3.2. Overall O_x -visit association

We observed a U-shaped crude association between O_x and visits, with a minimal risk at 32 ppb, as shown in Fig. 2. After adjusting for NO_2 (grey, triangular central estimates), changes, though apparent, were minimal and not statistically different from the O_x -only model (in 1st percentile, Figure S2). However, NO-adjustment (grey, square-shaped central estimates) resulted in a substantial and statistically significant (*p-value* < 0.001) reduction in the lower risk curve, suggesting the probable importance of NO exposure on cardiorespiratory clinic visits. While we observed an increasing risk with increasing concentration beyond the MRP, at the 99th percentile all pollutant-adjusted models were not statistically different from the O_x -only model (Fig S2). Visually, NO- or NO_2-adjusted O_x -cardiorespiratory associations exhibited a potential threshold model, with a linear increase beyond the MRP. However, the results from the threshold analysis indicated a lack of evidence for a threshold (Fig S5).

3.3. Season-specific O_x C-R associations

As shown in Fig. 3, the O_x -related risks in summer increased nearly linearly with increasing O_x concentrations. During winter, the O_x C-R function included a null association at the upper concentrations, with a departure from the null association at the lower concentrations. However, we noted a substantial reduction in the risks at the lower concentration range after adjustment for NO and NO_2, with a more evident reduction with NO, which approached the null association during winter. This is supported by the significant reduction in the lower risk curve (represented by the 1st percentile, as shown in Fig S3) after adjusting for NO (*p-value* $= 0.02$) during winter. We did not observe any difference between the crude model and the adjusted models for the 1st percentile in either summer or winter, or in the 99th percentile of winter.

4. Discussion

A crude O_x -cardiorespiratory visit exhibited a U-shaped pattern. However, adjustment of the oxides of nitrogen, particularly NO, attenuated the lower risk curve and subsequently altered the shape of the crude O_x C-R function. Season-specific analysis revealed the relative importance of specific oxides of nitrogen in the attenuation of the O_x C-R function.

4.1. Non-linearity of the crude O_x-health association

In the current study, the crude O_x-cardiorespiratory association has showed an indication of a potential threshold around 32 ppb, similar to

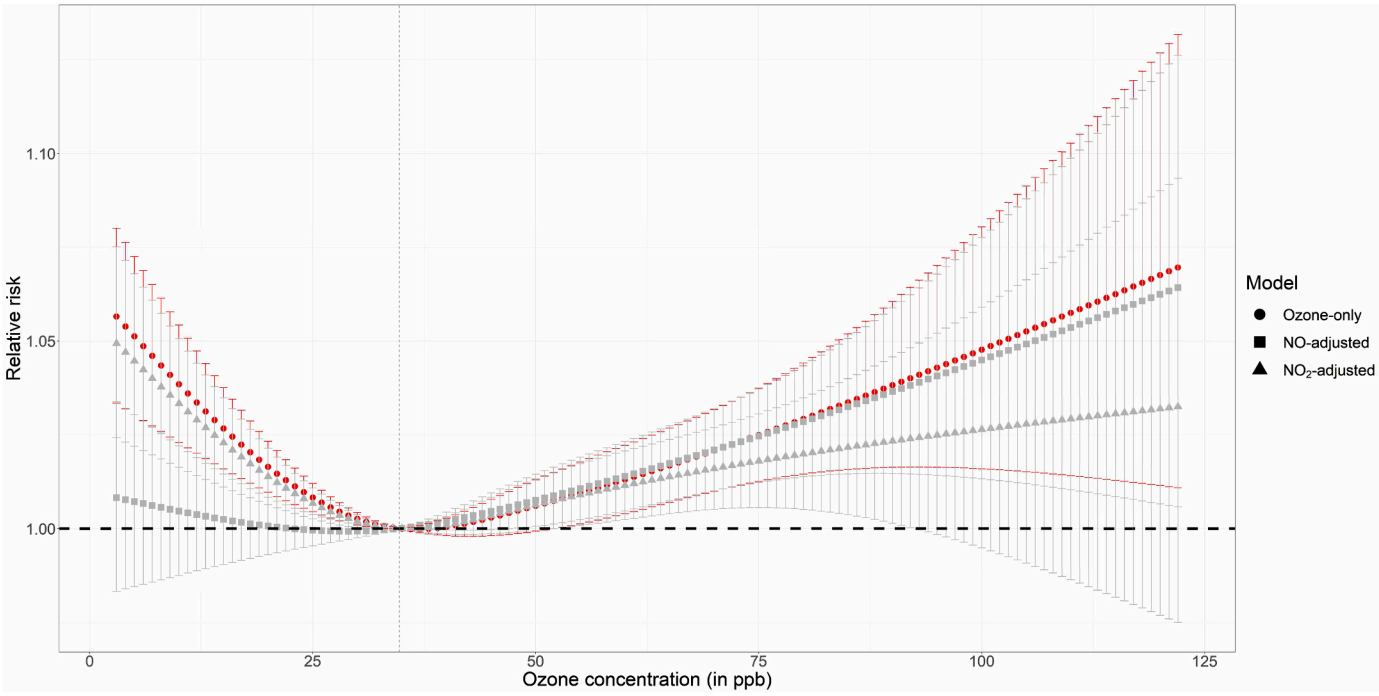


Fig. 2. Pooled, year-round ozone-cardiorespiratory associations, Ozone-only model is represented by solid, red circles, whereas NO- and NO_2-adjusted models are shown in grey square and triangles, respectively. Vertical dashed line indicates the location of the candidate threshold at 32 ppb. Horizontal dashed line indicates the null association when relative risk is at 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

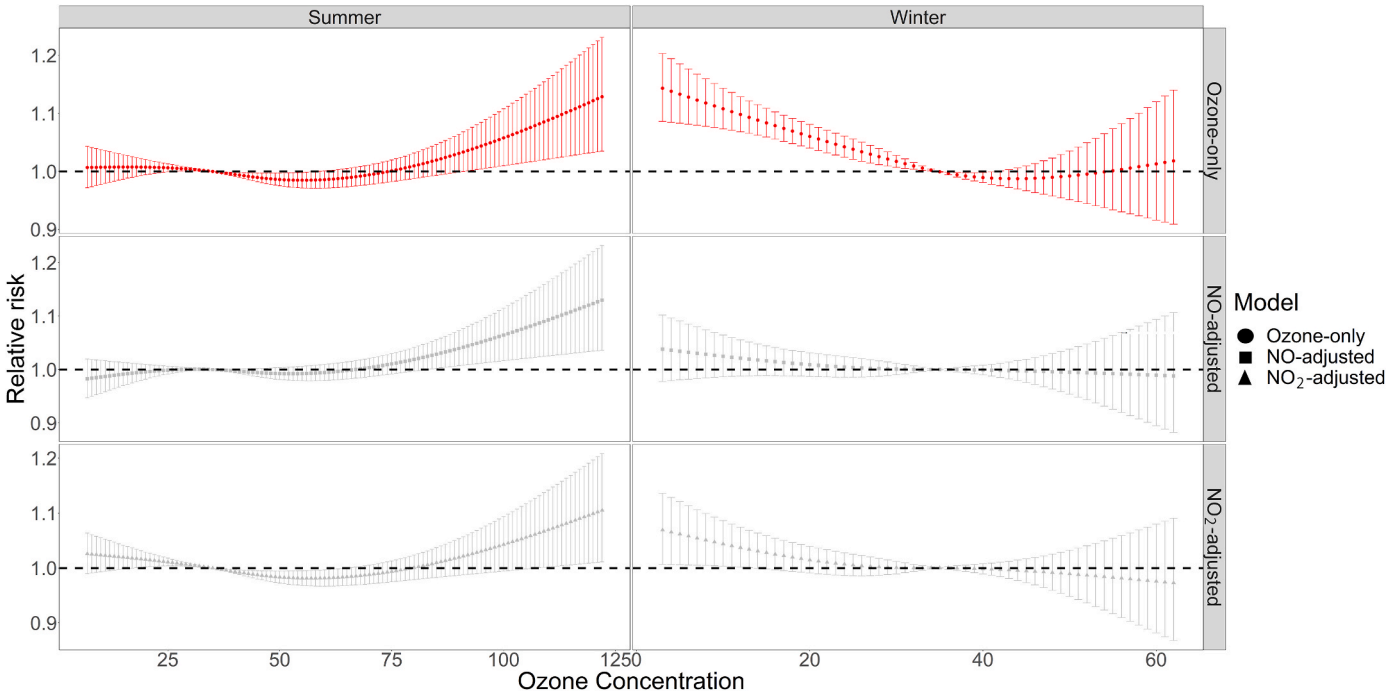


Fig. 3. Pooled, season-round ozone-cardiorespiratory associations, Ozone-only model is represented by solid, red circles, whereas NO- and NO_2-adjusted models are shown in grey square and triangles, respectively. Horizontal dashed line indicates the null association when relative risk is at 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

previous studies (Bae et al., 2015; Faustini et al., 2019; Huang et al., 2019; Wang et al., 2020). Bae et al. (2015) noted a threshold point, or in this case, the MRP, of approximately 30–40 ppb for 13 Japanese and Korean cities. In the USA, Stylianou and Nicolich (2009) similarly observed non-linear O_3 and all-cause mortality associations with thresholds ranging from 21 to 44 ppb. Across the non-linear risk curve, we noted substantial statistical uncertainties at the extremes, with wider confidence intervals in the upper extreme concentrations than in the lower extreme concentrations. These wide uncertainties at the extremes are potentially due to sparse data, which have also been observed in other studies (Bae et al., 2015; Faustini et al., 2019).

There is no clear physiological mechanism, however, which may explain how low O_x concentrations, particularly those approaching zero would lead to an increase in the risk. This leads to an inquiry as to whether the estimated incremental risk with decreasing O_x concentration is a true association or an artefact of the limited modeling flexibility in covariate adjustment across the exposure range. After adjusting for several C-R functions, we found that adjustment for either NO or NO_2 led to a substantial reduction in the O_x-visit risk curve, particularly in the lower concentration range (Fig. 2). The observed reduction may be related to the chemical mechanism, specifically, the photochemical process involved in the O_x and oxides of nitrogen titration.

4.2. Photochemical processes of O_x alongside NO and NO_2

Photochemical processes involved in the production of tropospheric O_3 include precursors such as NO and NO_2, as well as volatile organic compounds (VOCs). NO_2 and NO, in particular, titrate O_3, hence in high NO_2 and NO concentrations, O_3 is depleted (Han et al., 2011; Song et al., 2011). More detailed photochemical processes have been discussed previously (Crutzen, 1979; Wu et al., 1973). A large body of evidence has linked oxides of nitrogen exposure to various physiological inflammatory responses (Persinger et al., 2001; Ricciardolo et al., 2004; van der Vliet et al., 2000), and which may lead to the development of short-/long-term morbidity outcomes (Gillespie-Bennett et al., 2011; Studnicka et al., 1997), and even death (Faustini et al., 2014; Samoli et al., 2006). The high concentration of the oxides of nitrogen, during the low O_x concentration period, together with the documented health risks, posits a candidate mechanism, which may partially explain the observed health risks at low O_x concentrations. This is further highlighted in the following section in relation to the season-specific analysis.

4.3. O_x C-R functions by season

In Fig. 3, we observed a generally increasing risk during summer and a weakly decreasing risk during winter. Several studies have also noted similar yet inconsistent trends of increasing risks during summer and statistically significant and non-significant reduced risks during winter (Chen et al., 2014; Zanobetti and Schwartz, 2008). Such trends may be related to photochemical processes leading to different exposure range and thus a difference in the risks, as mentioned in the earlier section. It is noteworthy that the adjustment of NO or NO_2 did not substantially change the crude O_x C-R function during summer. However, this was not the case for winter, when an apparent reduction in the risk, especially in the lower part of the crude O_x C-R function, was observed. Williams et al. (2014) noted that changes in O_3-related mortality risks after adjusting for NO_2 were more apparent in winter than in summer, in London. On the contrary, Bero Bedada et al. (2016) noted a larger risk reduction in summer than in winter after NO_2 adjustment in Stockholm County. To date, there is no conclusive evidence on the direction of the risks related to the oxides of nitrogen-adjusted O_x-health outcomes during summer and winter, and thus this remains to be further elucidated in future studies.

4.4. Impact of NO on the O_x C-R association

It is noteworthy that the NO-adjustment led to an even greater attenuation of the lower risk curve than the NO_2-adjustment, as shown in Figs. 2 and 3. The adjustment diminished the lower risk curve, which crossed the null association below the MRP, while maintaining a positive and near-linear increase in risk above the MRP. There is no existing

plausible pathophysiological mechanism which could explain, at the current time, why there is a difference in the risk, however, we presume that this may be due to the difference in the C-R curves between NO and NO_2.

As shown in Fig S4 (upper panel), both the NO and NO_2 C-R functions have similar and overlapping functional shapes. However, there is a notable divergence between the two C-R functions beyond the 96th percentile (Fig S4, lower panel), with NO having a higher risk than NO_2. These results suggest that the higher risks of NO over NO_2 in the extreme concentration range, if adjusted accordingly, would correspond to the difference in the magnitude of reduction observed in the lower O_x concentration range for either NO or NO_2, as previously shown in Fig. 2 or in the 1st percentile of Fig S2.

In the subsequent sensitivity analysis, we examined the impact of the oxides of nitrogen on the lagged O_x exposure-response associations (as shown in Figs S7 and S8). The impact of NO and NO_2 was limited to the same-day association (lag 0) and exhibited minimal to no impact from the subsequent lags from lag 1 to 7. The immediate effect of the oxides of nitrogen adjustment on the O_x exposure-response association may be related to the titration process between nitrogen and O_x, which occurs over a short period of time.

4.5. Linearity of the O_x-cardiorespiratory visit: no evidence of a threshold

The lowest potential threshold was 32 ppb (Fig S5). The potential threshold (at 32 ppb) offered a marginally better model fit than the no-threshold model, with a meager difference of 0.21 between the two models. However, because the difference was less than 2, compelling evidence of a threshold is lacking in this case. This concurs with previous studies whereby in contiguous 98 US urban communities, Bell et al. (2006) noted that amidst the use of a spline approach, there was a monotonic linear increase in the risk from the minimum concentration range towards the uppermost exposure range. Chen et al. (2017) shares a similar observation in Jiangsu, China, where the potential threshold at 90 ppb offered a minimally better fit than the no-threshold model, but ultimately within a 2 qAIC difference.

5. Limitations

Although we have employed rigorous statistical modeling techniques to adjust for the relevant covariates and conducted several sensitivity analyses to establish the robustness of our results, there are still several key limitations to which succeeding studies can refer and build upon. The current study is limited to clinic visit information aggregated at the city level, which may have affected exposure matching. We crudely assumed that these clinic visits occur in populated areas in the city where these monitoring stations are co-located. However, with the lack of granular data, we cannot fully ascertain this assumption; hence, there is a need for more detailed address information whenever available. More detailed data in terms of sex, age and cause-specific categories would provide deeper insights into the role of possible effect modifiers. Concurrent with the limited number of cities is a generalization of the results. Caution should be exercised with the interpretation of results, particularly regarding the generalization to other cities across the country. After a sensitivity analysis adjusting for particulate matter with a size of 2.5 μm or less in diameter ($PM_{2.5}$), the risk curve remained unchanged (Fig S6). However, there may be other pollutants or aerosols that may impact the C-R curve due to data unavailability; these effects could not be explored. Finally, we offer only a partial explanation of the difference in the C-R functions for NO and NO_2 from a data-driven perspective. It is reasonable to believe that the functional C-R shapes are driven by differential toxicity, especially at the upper extreme concentrations. Future studies could benefit from relevant toxicological studies (i.e. mouse models) to explore this phenomenon.

6. Conclusions

After adjusting for NO, the risk of cardiorespiratory visits after O_x exposure increased linearly with no apparent evidence of a threshold. Contrary to expectations, NO adjustment led to a statistically significant reduction in O_x-cardiorespiratory risks in the low concentration range compared with NO_2-adjustment. The observed differences after the respective adjustments and their impact on the O_x-cardiorespiratory C-R curve may, in part, be related to the differential toxicity of the exposures. Further investigation of the role of differential toxicity, in relation to NO and NO_2 exposure is required to elucidate these results.

Credit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2022.120802.

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